Local DNS Lab

By Keenan Morrison

First I created the debian template by installing the debian template to proxmox

```
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@pve1:~# ls
root@pve1:~# pveam update
update successful
root@pve1:~# pveam available --section system | grep debian-13
system          debian-13-standard_13.1-2_amd64.tar.zst
root@pve1:~# pveam download  debian-13-standard_13.1-2_amd64.tar.zst
400 not enough arguments
pveam download <storage> <template>
root@pve1:~# pveam download  local debian-13-standard_13.1-2_amd64.tar.zst
downloading http://download.proxmox.com/images/system/debian-13-standard_13.1-2_amd64
.tar.zst to /var/lib/vz/template/cache/debian-13-standard_13.1-2_amd64.tar.zst
--2026-05-11 02:37:01--  http://download.proxmox.com/images/system/debian-13-standard
_13.1-2_amd64.tar.zst
Resolving download.proxmox.com (download.proxmox.com)... 66.70.154.82, 2607:5300:400:
7d00::80
```

Figure 1.1

Then I added the hostname pihole and a complex password that I uploaded to a offline password manager

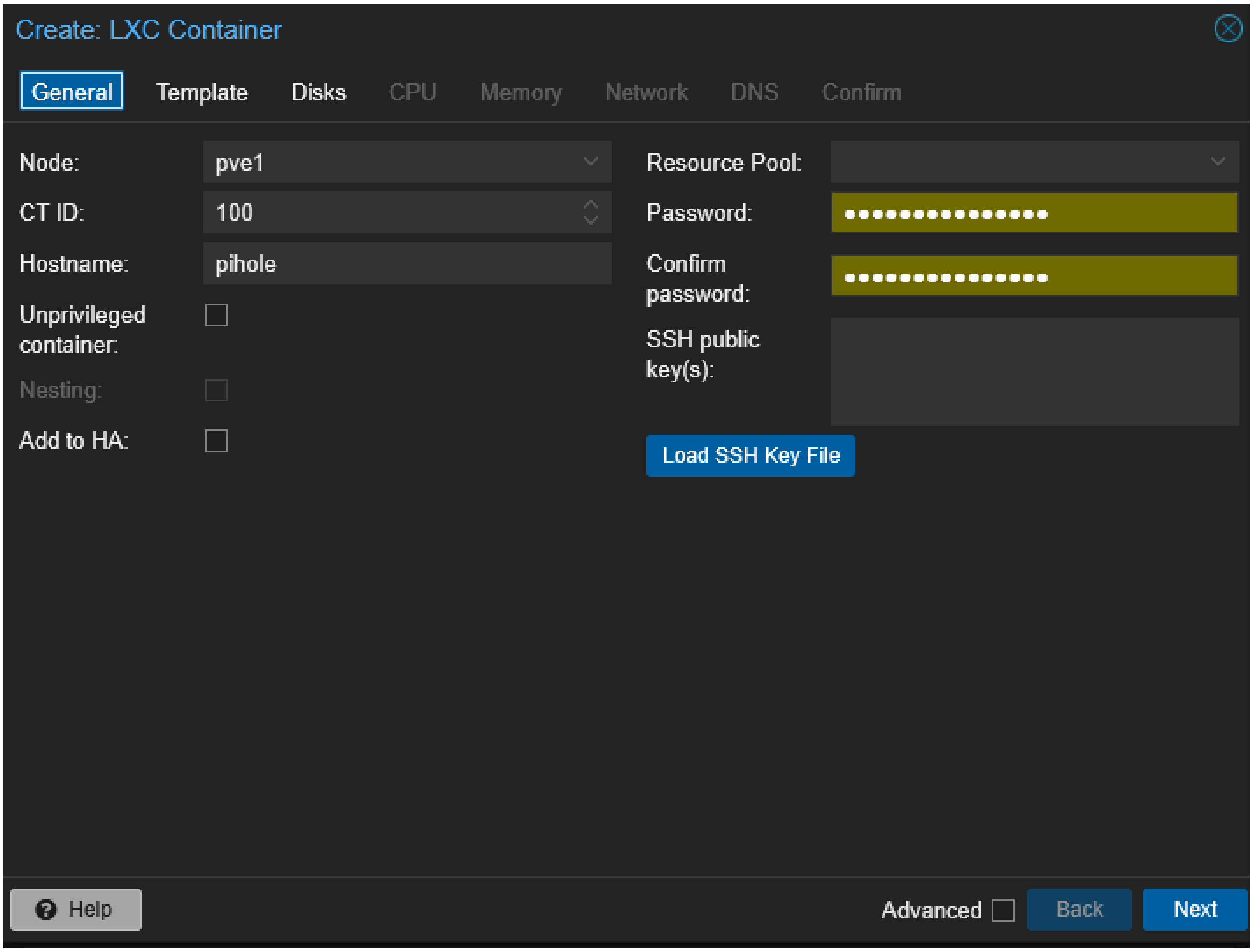

Figure 1.2

I added the template Debian 13 that I created earlier

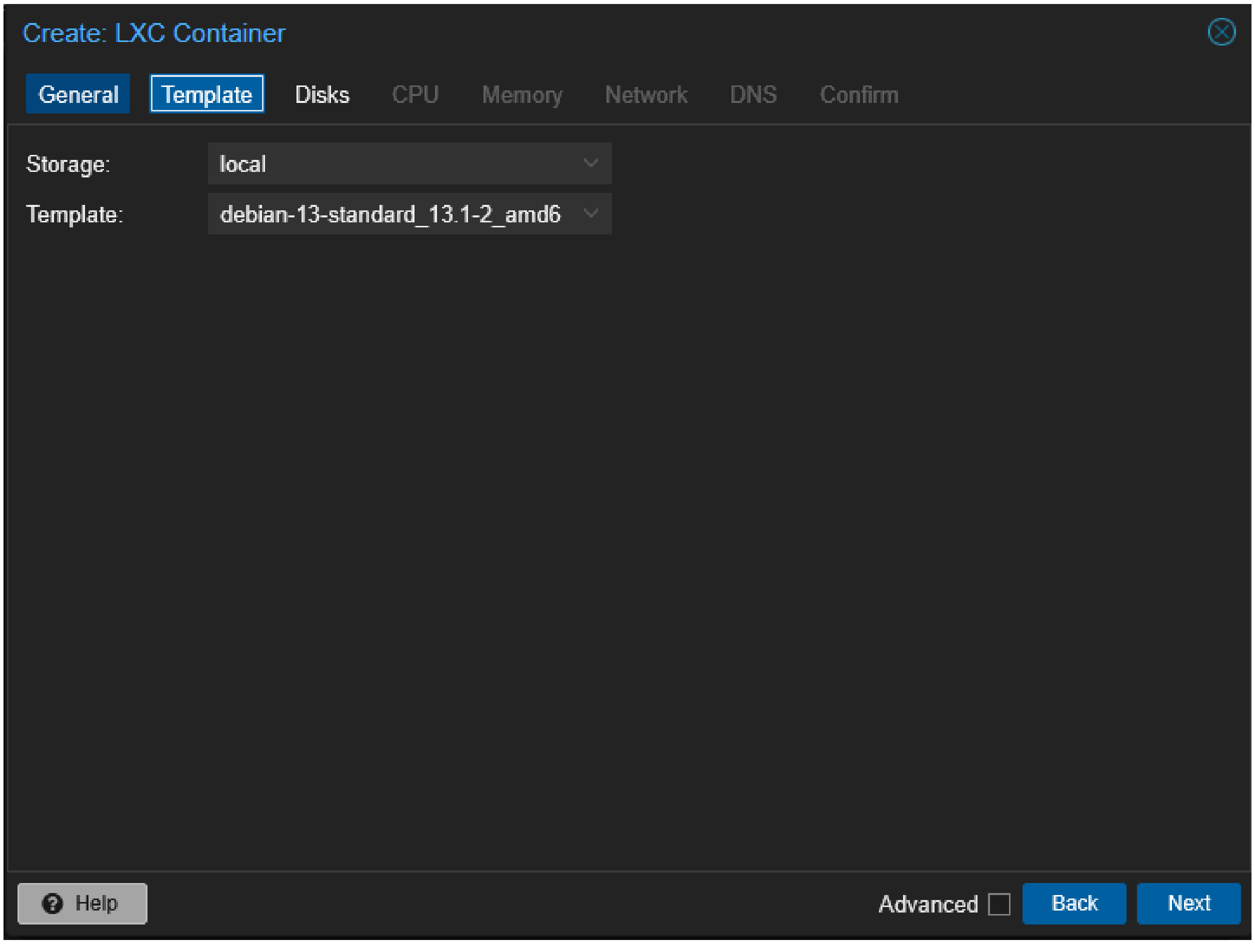

Figure 1.3

Then I configured the CPU , Memory , Disks for the LXC container and it should look like this.

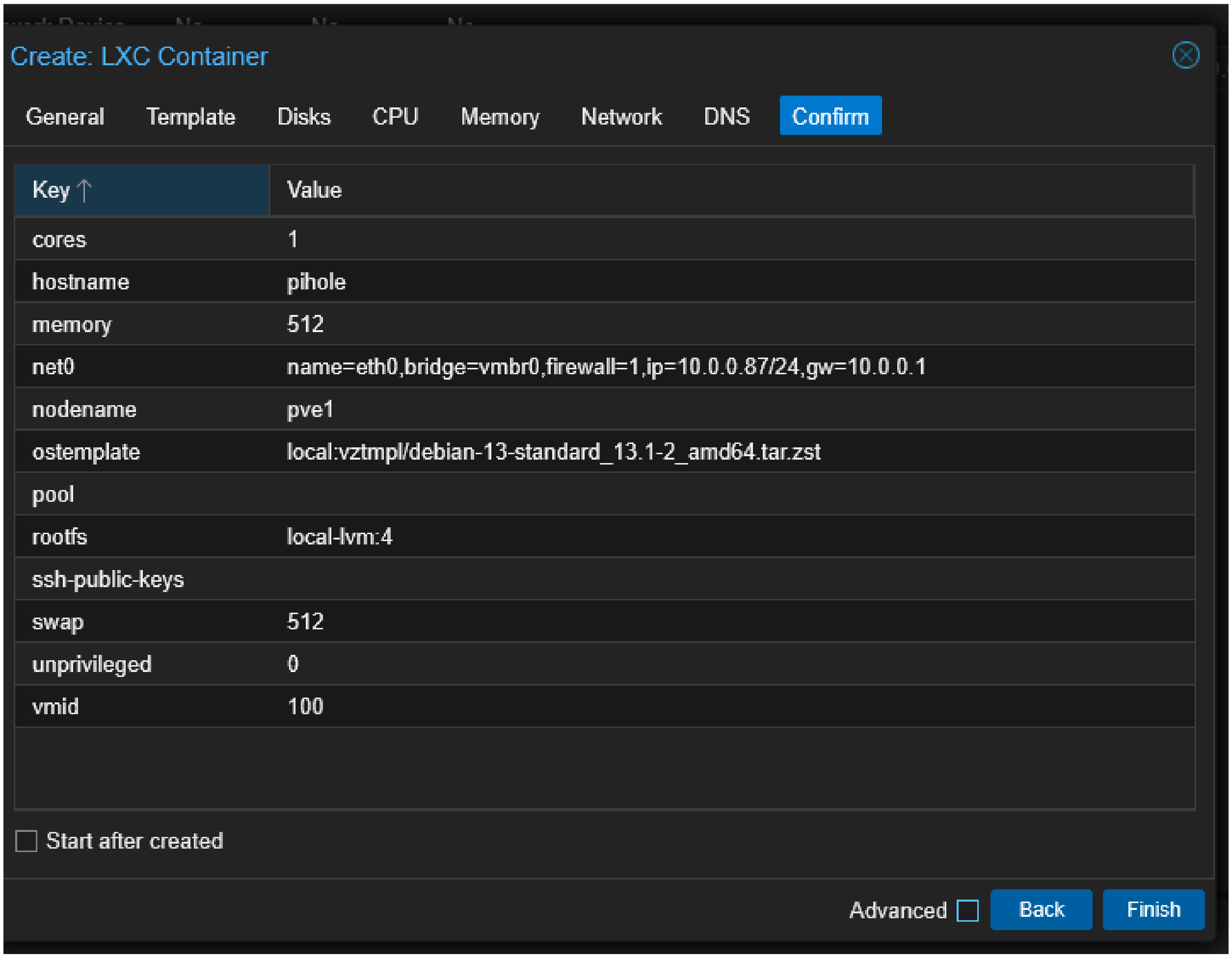

Figure 1.4

The container ID is 100 as can be seen by this screenshot

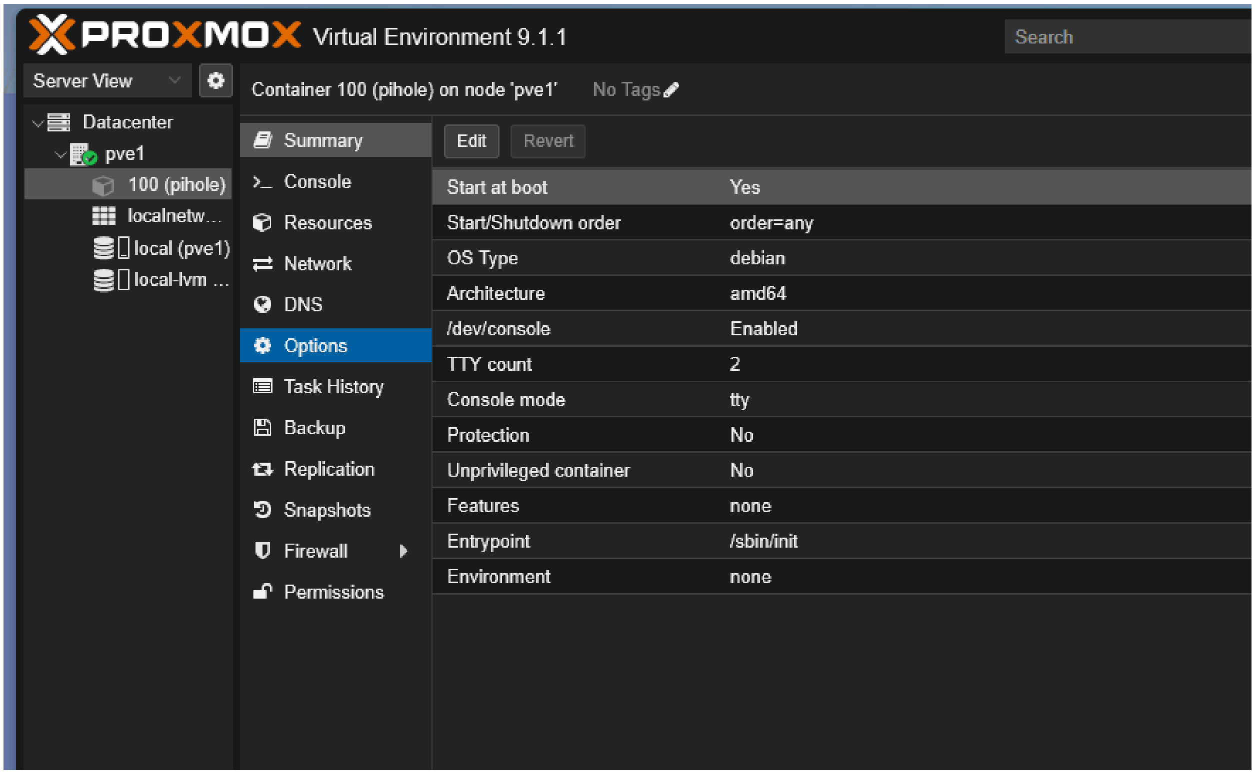

Figure 1.5

Then I went to pve1 and entered the command **pct enter 100**. After this I started the pihole script and this is what you should see.

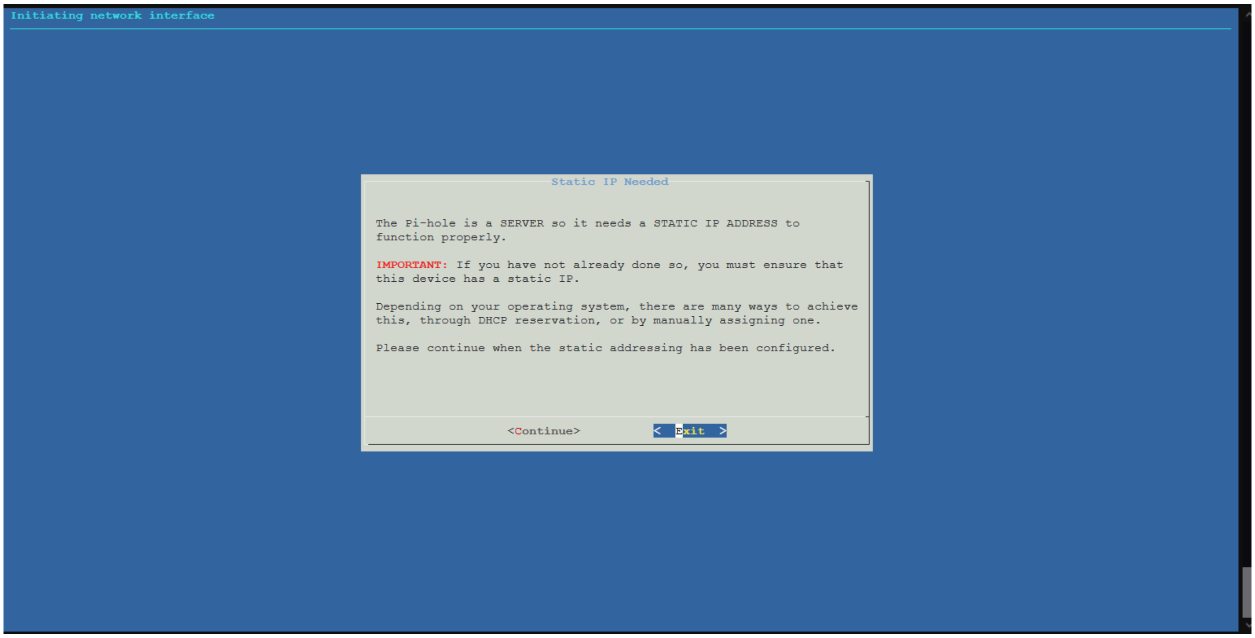

Figure 1.6

For the DNS I chose Cloudflare because I personally like it

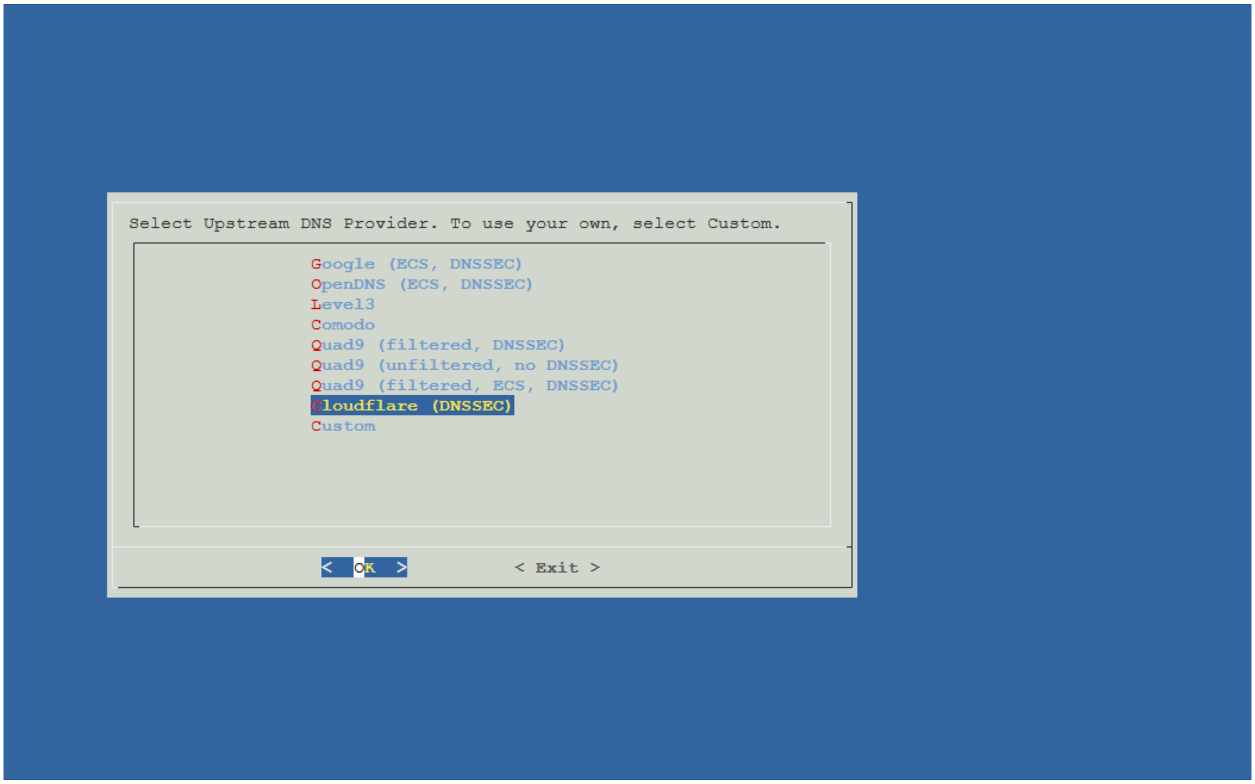

Figure 1.7

After that you should go to http://IP_ADDRESS/admin you configure and you should see this dashboard

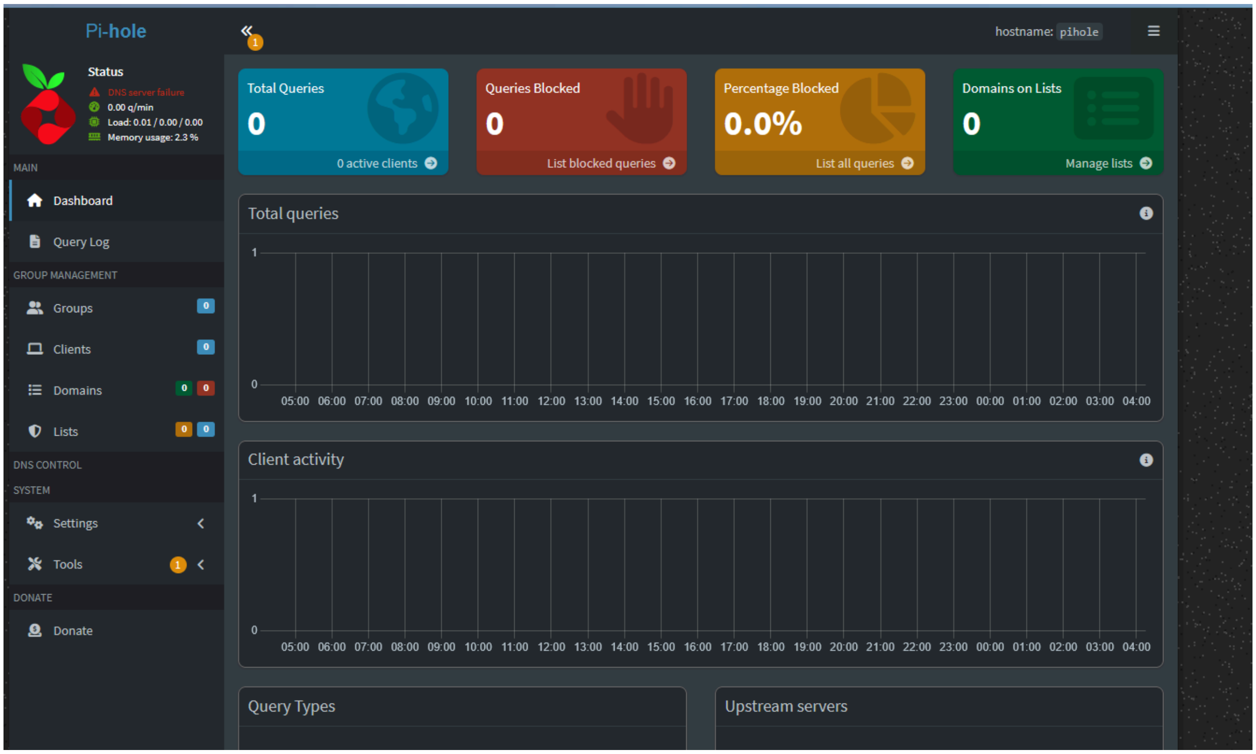

Figure 1.8

Then click Settings > Local DNS Records. It should look like the image shown in. When you are here then press the green plus button and add the ip of a locally hosted domain in the “Associated IP” section and for the domain put your desired domain name.

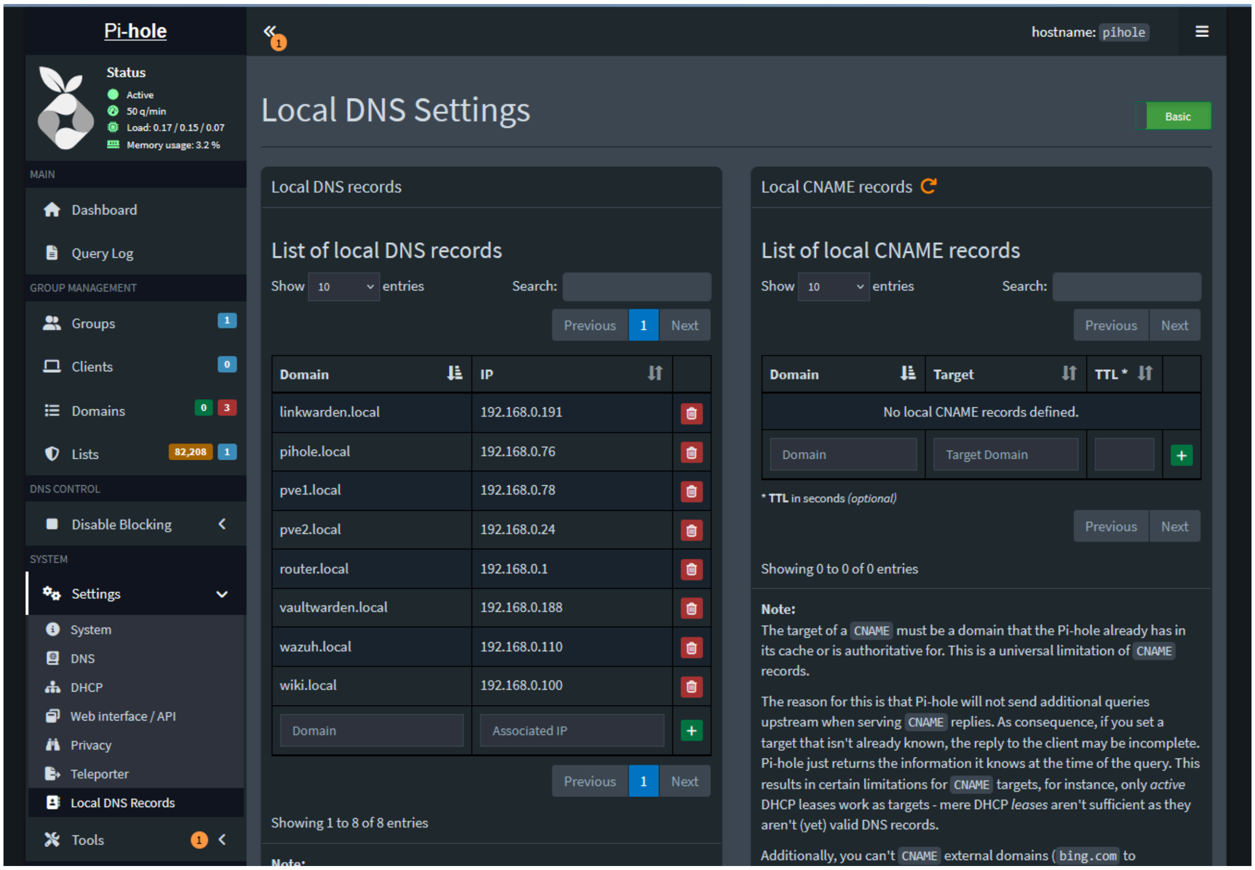

Figure 1.9

Overview

In this lab I set up a program called Pihole and configured it to be a locally hosted DNS server that can help me block ads and configure local DNS records to access services in my network a lot easier by committing words to memory instead of IPs.